1. Which article is used before a singular countable noun with a consonant sound?

(A) An
(B) A
(C) The
(D) Zero article

2. Which article correctly precedes the acronym 'HPLC' (pronounced aitch-pee-el-see)?

(A) A
(B) An
(C) The
(D) No article

3. Which article correctly precedes 'European clinical trial'?

(A) An
(B) A
(C) The
(D) Zero article

4. Why does 'European' take the indefinite article 'a'?

(A) It starts with vowel E
(B) It begins with consonant glide /j/ sound
(C) It is a proper adjective
(D) It is capitalized

5. Choose the correct article: 'The enzyme was incubated for ___ hour.'

(A) a
(B) an
(C) the
(D) no article

6. Which article correctly precedes 'RNA molecule' (pronounced ar-en-ay)?

(A) A
(B) An
(C) The
(D) Zero article

7. The definite article 'the' is used to refer to:

(A) Any random entity
(B) A specific or unique entity
(C) Plural nouns only
(D) Uncountable nouns only

8. Identify the sentence that correctly omits articles before a generic mass noun:

(A) The DNA is essential for life.
(B) A DNA is essential for life.
(C) DNA is essential for life.
(D) An DNA is essential for life.

9. Choose the correct article: 'We purchased ___ new spectrophotometer for the lab.'

(A) a
(B) an
(C) the
(D) zero article

10. In 'We bought a centrifuge; ___ centrifuge was placed on bench 3', the missing word is:

(A) a
(B) an
(C) the
(D) some

11. The transition from 'a centrifuge' to 'the centrifuge' is an example of:

(A) Cataphoric reference
(B) Anaphoric reference
(C) Exophoric reference
(D) Elliptical reference

12. Which of the following nouns takes ZERO article in generic academic statements?

(A) Biotechnology
(B) The Biotechnology
(C) A Biotechnology
(D) An Biotechnology

13. Choose the correct determiner: '___ student must wear personal protective equipment.'

(A) All
(B) Every
(C) Both
(D) Many

14. Which quantifier is used exclusively with uncountable mass nouns?

(A) Many
(B) Much
(C) Few
(D) Several

15. Which quantifier is used exclusively with countable plural nouns?

(A) Much
(B) Many
(C) Little
(D) Less

16. The phrase 'a few' signifies:

(A) Almost none (negative connotation)
(B) Some / a small positive quantity
(C) An excessive number
(D) Zero quantity

17. The bare quantifier 'few' without 'a' signifies:

(A) A positive quantity
(B) Scarcity / almost none (negative)
(C) An abundant quantity
(D) A definite quantity

18. Choose the correct quantifier: 'We have ___ time left before the thermal cycler stops.'

(A) few
(B) little
(C) many
(D) several

19. Choose the correct quantifier: 'There were ___ colonies observed on the negative control plate.'

(A) little
(B) few
(C) much
(D) less

20. What type of determiner is 'these' in: 'These cultures grew well'?

(A) Possessive
(B) Demonstrative
(C) Interrogative
(D) Distributive

21. Identify the possessive determiner among the following words:

(A) Their
(B) Theirs
(C) Them
(D) They

22. Choose the correct word: 'The plasmid lost ___ ability to replicate under high temperature.'

(A) it's
(B) its
(C) its'
(D) it

23. Why is 'it's' incorrect as a possessive determiner?

(A) 'It's' is a contraction for 'it is'
(B) It's is plural
(C) It's is archaic
(D) It's is an adverb

24. Which of the following takes the definite article 'the'?

(A) Mount Everest
(B) Lake Superior
(C) Pacific Ocean
(D) Asia

25. Which country name requires the definite article 'the'?

(A) France
(B) Germany
(C) Netherlands
(D) India

26. In 'The more you vortex, ___ more foam is generated', the missing word is:

(A) a
(B) the
(C) an
(D) some

27. Choose the correct determiner: '___ of the two candidate vaccines proved effective.'

(A) Both
(B) All
(C) Every
(D) Each other

28. Choose the correct determiner: '___ of the three candidates was accepted.'

(A) Neither
(B) None
(C) Both
(D) Either

29. In 'A single bacterium divided', 'A' functions as:

(A) An indefinite article / numerical determiner
(B) A definite article
(C) A demonstrative pronoun
(D) A coordinating conjunction

30. Choose the correct article: 'He is ___ honest researcher.'

(A) a
(B) an
(C) the
(D) zero article

31. Why does 'honest' take 'an'?

(A) It begins with vowel sound /o/ (silent h)
(B) It has six letters
(C) It is an adjective
(D) It is of French origin

32. Choose the correct article: 'She has ___ PhD in molecular genetics.'

(A) a
(B) an
(C) the
(D) no article

33. Choose the correct article: 'She holds ___ MSc degree in biotechnology.'

(A) a
(B) an
(C) the
(D) no article

34. Why does 'MSc' take 'an'?

(A) Pronounced 'em-es-see', starting with vowel /e/
(B) M is a vowel
(C) It is a postgraduate degree
(D) It takes no article

35. Choose the correct determiner: '___ researcher submitted their data.'

(A) Each
(B) All
(C) Both
(D) Several

36. Identify the sentence that uses the definite article generically for an invention:

(A) The microscope transformed cell biology.
(B) A microscope was dropped.
(C) Microscopes are in room 2.
(D) One microscope was calibrated.

37. Choose the correct determiner: 'We found ___ contamination in the sterile laminar airflow.'

(A) no
(B) not
(C) none
(D) neither

38. In 'Any unauthorized personnel will be reported', 'Any' functions as:

(A) A demonstrative determiner
(B) An indefinite quantifier
(C) A possessive determiner
(D) A relative pronoun

39. Which phrase correctly expresses frequency with an indefinite article?

(A) Once per a day
(B) Twice a week
(C) Three times the month
(D) Four times in week

40. Choose the correct article: '___ oxygen is required for cellular respiration.'

(A) An
(B) The
(C) Zero article
(D) A

41. Identify the sentence with correct article usage for a superlative adjective:

(A) Taq is most thermostable enzyme.
(B) Taq is the most thermostable enzyme.
(C) Taq is a most thermostable enzyme.
(D) Taq is most thermostable.

42. Select the sentence with correct article usage for academic disciplines:

(A) She studies the Molecular Biology.
(B) She studies a Molecular Biology.
(C) She studies Molecular Biology.
(D) She studies an Molecular Biology.

43. In 'The enzyme has ___ affinity for the substrate, so binding was negligible':

(A) little
(B) a little
(C) few
(D) a few

44. In 'We detected ___ residual activity, allowing us to proceed with the assay':

(A) little
(B) a little
(C) few
(D) a few

45. Choose the correct determiner: 'There were ___ colonies than expected on the plate.'

(A) less
(B) fewer
(C) lesser
(D) fewest

46. Choose the correct determiner: 'The reaction required ___ energy than anticipated.'

(A) fewer
(B) less
(C) few
(D) lesser

47. Identify the sentence with correct article usage before musical instruments vs scientific equipment:

(A) The student plays the piano and uses the spectrophotometer.
(B) The student plays a piano and uses a spectrophotometer.
(C) The student plays piano and uses spectrophotometer.
(D) The student plays the piano and uses spectrophotometer.

48. Select the sentence with correct article usage for diseases:

(A) He was diagnosed with the malaria.
(B) He was diagnosed with malaria.
(C) He was diagnosed with a malaria.
(D) He was diagnosed with an malaria.

49. Which disease traditionally takes the definite article 'the' in colloquial English?

(A) The flu (influenza)
(B) The cancer
(C) The diabetes
(D) The tuberculosis

50. Choose the correct article: 'We conducted ___ in vitro assay.'

(A) a
(B) an
(C) the
(D) no article

51. Choose the correct article: 'We performed ___ in vivo validation study.'

(A) a
(B) an
(C) the
(D) no article

52. Identify the sentence that correctly avoids an unattended demonstrative pronoun:

(A) This led to catastrophic cell death.
(B) This error led to catastrophic cell death.
(C) This was because the temperature rose.
(D) This indicated that yield was low.

53. Choose the correct article: '___ University of Oxford published the trial results.'

(A) A
(B) An
(C) The
(D) Zero article

54. Choose the correct article: 'She graduated from ___ Stanford University.'

(A) a
(B) an
(C) the
(D) zero article

55. Why does 'Stanford University' take zero article while 'University of Oxford' takes 'The'?

(A) 'University of...' has an 'of'-phrase postmodifier requiring 'The'
(B) Stanford is older
(C) Oxford is British
(D) Stanford is private

56. Choose the correct determiner: '___ of the five samples displayed identical band migration.'

(A) All
(B) Both
(C) Neither
(D) Either

57. Choose the correct determiner: '___ researcher brought his own calibrated pipette.'

(A) Each
(B) All
(C) Both
(D) Several

58. Identify the sentence containing an incorrect article:

(A) The water boils at 100 deg C under standard atmospheric pressure.
(B) Water boils at 100 deg C under standard atmospheric pressure.
(C) The water in this beaker boils at 98 deg C.
(D) Purified water was added to the mixture.

59. Why is 'The water boils at 100 deg C...' incorrect in A?

(A) Water is used generically as a mass noun and requires zero article
(B) Boils is singular
(C) 100 deg C needs an article
(D) Pressure is uncountable

60. Choose the correct article: '___ Nile River flows through Egypt.'

(A) A
(B) An
(C) The
(D) Zero article

61. Choose the correct determiner: 'We evaluated ___ candidate vector, but neither replicated.'

(A) either
(B) neither
(C) both
(D) each

62. In 'Neither vector replicated', 'Neither' functions as:

(A) A negative disjunctive determiner
(B) A coordinating conjunction
(C) A relative pronoun
(D) An adverb

63. Select the sentence with correct article usage before unique celestial entities:

(A) The Moon orbits Earth.
(B) The Moon orbits the Earth.
(C) Moon orbits the Earth.
(D) A Moon orbits a Earth.

64. Choose the correct article: 'Dr. Jones was elected ___ chairperson of the bioethics panel.'

(A) a
(B) an
(C) the
(D) zero article

65. Why does 'chairperson' take zero article after verbs like 'elect' or 'appoint'?

(A) It is a unique office/title functioning as objective complement
(B) It is a mass noun
(C) Chairperson is non-count
(D) It is an adjective

66. Select the sentence showing correct use of 'such a' vs 'such':

(A) We encountered such a severe contamination that cultures died.
(B) We encountered such severe contamination that cultures died.
(C) Both A and B are grammatically valid depending on countability.
(D) Neither A nor B is correct.

67. Choose the correct determiner: '___ cellular organisms possess ribosomes.'

(A) Every
(B) All
(C) Each
(D) One

68. Choose the correct article: '___ United Kingdom has strict gene-editing guidelines.'

(A) A
(B) An
(C) The
(D) Zero article

69. Choose the correct article: '___ India has expanded its biopharmaceutical manufacturing base.'

(A) A
(B) An
(C) The
(D) Zero article

70. In 'We observed a decrease in cell size', 'a' is required because:

(A) 'Decrease' is here used as a singular countable nominal event
(B) 'Decrease' is always uncountable
(C) 'Decrease' is a verb
(D) 'Cell size' is plural

71. Choose the correct determiner: 'Do you require ___ further assistance with the chromatography?'

(A) any
(B) some
(C) many
(D) few

72. In affirmative offers or requests expecting agreement, standard English uses:

(A) Any
(B) Some
(C) None
(D) Few

73. Choose the correct article: '___ richer the culture medium, ___ faster the microbial doubling.'

(A) The / the
(B) A / a
(C) An / an
(D) Zero / zero

74. Select the sentence that correctly avoids article repetition before a dual-role noun:

(A) The director and the principal investigator was present.
(B) The director and principal investigator was present.
(C) The director and principal investigator were present.
(D) A director and a principal investigator was present.

75. In 'The director and principal investigator was present', one article signifies that:

(A) Both titles refer to the same individual
(B) Two distinct individuals were present
(C) The director is absent
(D) The sentence is ungrammatical

76. Identify the sentence with an erroneous article insertion in an idiomatic scientific phrase:

(A) The cells were analyzed in the vitro.
(B) The cells were analyzed in vitro.
(C) The reaction reached at an equilibrium.
(D) Both A and C contain erroneous article insertions.

77. Select the sentence with correct article usage regarding historical periods:

(A) During the Middle Ages, alchemy flourished.
(B) During Middle Ages, alchemy flourished.
(C) During a Middle Ages, alchemy flourished.
(D) In Middle Ages, alchemy flourished.

78. Choose the correct article: 'The patient was admitted to ___ hospital for immunotherapy.'

(A) a
(B) an
(C) the
(D) zero article (British) / the (American)

79. In 'The protein has ___ affinity for ligand A and ___ affinity for ligand B':

(A) a high / a low
(B) high / low
(C) the high / the low
(D) Both A and B are acceptable in scientific style

80. Select the sentence with flawless article usage in a complex technical description:

(A) The synthesis of a recombinant antibody requires an optimized host system and the precise regulation of pH.
(B) A synthesis of the recombinant antibody requires optimized host system and precise regulation of the pH.
(C) The synthesis of recombinant antibody requires the optimized host system and precise regulation of pH.
(D) Synthesis of the recombinant antibody requires an optimized host system and the pH regulation.

81. Choose the correct determiner: '___ member of the sequencing consortium signed the data-sharing pact.'

(A) Every
(B) All
(C) Both
(D) Several

82. Identify the sentence that violates determiner-noun concord:

(A) These species of bacteria is pathogenic.
(B) This species of bacteria is pathogenic.
(C) These kinds of assays are sensitive.
(D) This data was statistically significant.

83. Why is 'This data was statistically significant' considered non-standard in formal scientific style?

(A) 'Data' is strictly a plural noun in traditional academic style, requiring 'These data were'
(B) 'Data' is an adjective
(C) 'Statistically' is misplaced
(D) 'Significant' requires an article

84. Choose the correct article: 'The sample was analyzed by ___ gas chromatography.'

(A) a
(B) an
(C) the
(D) zero article

85. Why do analytical techniques like 'gas chromatography' take zero article?

(A) They are treated as uncountable nominal fields / methodological processes
(B) They are proper nouns
(C) They take indefinite articles only
(D) They are foreign words

86. Select the sentence with correct article usage for chemical compounds:

(A) The glucose is converted into the pyruvate.
(B) Glucose is converted into pyruvate.
(C) A glucose is converted into a pyruvate.
(D) The glucose is converted into pyruvate.

87. Identify the sentence that correctly introduces a newly defined scientific term:

(A) We term this novel organelle 'the peroxisome'.
(B) We term this novel organelle 'a peroxisome'.
(C) We term this novel organelle 'peroxisome'.
(D) We term this novel organelle the 'a peroxisome'.

88. Choose the correct determiner: 'There is ___ evidence supporting the oncogenic role of this mutation.'

(A) compelling
(B) a compelling
(C) many
(D) several

89. Why does 'evidence' reject the indefinite article 'a' (*a compelling evidence)?

(A) 'Evidence' is strictly an uncountable mass noun in English
(B) 'Compelling' starts with a consonant
(C) 'Evidence' is already plural
(D) 'Evidence' requires 'the'

90. Select the sentence that properly uses the generic definite article for a class of organs:

(A) The heart pumps blood through the vascular system.
(B) Heart pumps blood through vascular system.
(C) A heart pumps blood through a vascular system.
(D) The hearts pump blood through the vascular system.

91. Choose the correct determiner: 'The researcher spent ___ of his grant on mass spectrometers.'

(A) most
(B) most of
(C) the most
(D) mostly

92. In 'Most of the reagents were expired', what determiner structure is used?

(A) Partitive determiner construction
(B) Demonstrative adjective
(C) Possessive pronoun
(D) Negative quantifier

93. Select the sentence with correct article usage for physical constants and laws:

(A) According to the Beer-Lambert Law, absorbance is proportional to concentration.
(B) According to Beer-Lambert Law, absorbance is proportional to concentration.
(C) According to a Beer-Lambert Law, absorbance is proportional to concentration.
(D) According to the Beer-Lambert Law, the absorbance is proportional to a concentration.

94. Choose the correct article: 'The enzyme exhibited ___ Km of 0.4 mM.'

(A) a
(B) an
(C) the
(D) zero article

95. Why does 'Km' (pronounced Kay-em) take 'a' rather than 'an'?

(A) It begins with the consonant sound /k/ ('Kay')
(B) It is an abbreviation
(C) It is capitalized
(D) It is a constant

96. Choose the correct article: 'The mutant had ___ s-shaped growth curve.'

(A) a
(B) an
(C) the
(D) zero article

97. Why does 's-shaped' take 'an'?

(A) Pronounced 'es-shaped', beginning with vowel sound /e/
(B) S is a consonant
(C) Curve is singular
(D) It takes no article

98. Select the sentence with flawless determiner and article parallelism in a comparative analysis:

(A) The greater the purity of the sample, the higher the sensitivity of the assay.
(B) Greater purity of sample, higher sensitivity of assay.
(C) A greater purity of the sample, the higher sensitivity of assay.
(D) The greater purity of a sample, a higher sensitivity of an assay.

99. Choose the correct determiner: 'We tested five colonies; ___ of them harbored the recombinant insert.'

(A) none
(B) neither
(C) no
(D) any

100. Identify the sentence that demonstrates impeccable academic usage of articles, determiners, and mass noun quantification:

(A) A comprehensive analysis of the sequencing data revealed that fewer than five percent of the amplified reads contained spurious insertions.
(B) A comprehensive analysis of sequencing datas revealed that less than five percent of amplified reads contained spurious insertions.
(C) Comprehensive analysis of the sequencing data revealed that fewer than five percent of the amplified read contained spurious insertions.
(D) The comprehensive analysis of sequencing data revealed that less than five percents of amplified reads contained spurious insertions.

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the phonetic principles governing the selection between indefinite articles 'a' and 'an' in technical scientific terminology.

Q2. Analyze the anaphoric, cataphoric, and exophoric functions of the definite article 'the' in scientific manuscripts.

Q3. Detail the rules governing Zero Article (omission of articles) with uncountable nouns, plural nouns, and academic disciplines.

Q4. Examine the generic use of articles: 'A/An', 'The', and 'Zero Article' in defining biological taxa and classes.

Q5. Analyze quantifier syntax: 'Few' vs. 'A Few' and 'Little' vs. 'A Little' in evaluating experimental outcomes.

Q6. Examine the demonstrative determiners 'This', 'That', 'These', and 'Those' and the perils of the unattended 'this'.

Q7. Detail the grammatical rules governing articles with proper nouns, geographical entities, and institutional bodies.

Q8. Analyze the comparative correlative construction ('The more... the more') in expressing biophysical proportional relationships.

Q9. Examine distributive determiners: 'Each', 'Every', 'Either', and 'Neither' in experimental replicates and factorial designs.

Q10. Analyze article usage before mass nouns that take post-modifying 'of'-phrases in scientific descriptions.

Q11. Examine determiner usage with 'Much', 'Many', 'A lot of', and 'Plenty of' across formal and informal scientific registers.

Q12. Analyze the syntax and semantics of 'All', 'Whole', and 'Entire' in biological descriptions.

Q13. Examine article usage before acronyms, abbreviations, and symbols (e.g., pH, UV, DNA, mRNA) in biotechnology.

Q14. Analyze the omission of articles in paired idioms, parallel structures, and tabular lab protocols.

Q15. Examine the definite article 'the' with inventions, scientific instruments, and physiological systems.

Q16. Analyze the syntax of 'Both', 'Neither', and 'Either' with dual biological samples and bilateral structures.

Q17. Examine article usage before names of diseases, viruses, bacteria, and clinical medical conditions.

Q18. Detail the grammatical rules of 'Some' versus 'Any' in scientific hypotheses, negative findings, and controls.

Q19. Analyze the use of articles with chemical elements, compounds, functional groups, and molecular complexes.

Q20. Examine article usage before numerical expressions, percentages, fractions, and laboratory orders of magnitude.

Q21. Discuss the subtle error of repeating versus omitting the definite article before coordinate nouns.

Q22. Analyze the syntax and semantics of 'A number of' versus 'The number of' in clinical trial reporting.

Q23. Examine article usage with unique administrative, institutional, and academic titles (e.g., PI, Director, Dean).

Q24. Analyze determiner syntax with 'Another', 'Other', 'The other', and 'Others' in comparative laboratory trials.

Q25. Examine the definite article 'the' with academic degrees, certifications, and professional honors.

Q26. Analyze article usage before words like 'Nature', 'Society', 'Space', and 'Earth' in scientific writing.

Q27. Discuss determiner placement before coordinate adjectives and noun compounds in assay descriptions.

Q28. Examine the hazards of overusing 'the' (article clutter) in scientific methodology sections.

Q29. Analyze the syntactic distribution of 'Enough' as a determiner, adjective, and adverb in bioengineering.

Q30. Examine common translation transfer errors in article usage by non-native scientific researchers.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [B]
Q2: [B]
Q3: [B]
Q4: [B]
Q5: [B]
Q6: [B]
Q7: [B]
Q8: [C]
Q9: [A]
Q10: [C]
Q11: [B]
Q12: [A]
Q13: [B]
Q14: [B]
Q15: [B]
Q16: [B]
Q17: [B]
Q18: [B]
Q19: [B]
Q20: [B]

Q21: [A]
Q22: [B]
Q23: [A]
Q24: [C]
Q25: [C]
Q26: [B]
Q27: [A]
Q28: [B]
Q29: [A]
Q30: [B]
Q31: [A]
Q32: [A]
Q33: [B]
Q34: [A]
Q35: [A]
Q36: [A]
Q37: [A]
Q38: [B]
Q39: [B]
Q40: [C]

Q41: [B]
Q42: [C]
Q43: [A]
Q44: [B]
Q45: [B]
Q46: [B]
Q47: [A]
Q48: [B]
Q49: [A]
Q50: [B]
Q51: [B]
Q52: [B]
Q53: [C]
Q54: [D]
Q55: [A]
Q56: [A]
Q57: [A]
Q58: [A]
Q59: [A]
Q60: [C]

Q61: [D]
Q62: [A]
Q63: [B]
Q64: [D]
Q65: [A]
Q66: [C]
Q67: [B]
Q68: [C]
Q69: [D]
Q70: [A]
Q71: [A]
Q72: [B]
Q73: [A]
Q74: [B]
Q75: [A]
Q76: [D]
Q77: [A]
Q78: [D]
Q79: [D]
Q80: [A]

Q81: [A]
Q82: [D]
Q83: [A]
Q84: [D]
Q85: [A]
Q86: [B]
Q87: [A]
Q88: [A]
Q89: [A]
Q90: [A]
Q91: [A]
Q92: [A]
Q93: [A]
Q94: [A]
Q95: [A]
Q96: [B]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult] [illegible] DETAILED SOLUTIONS FOR SUBJECTIVE QUESTIONS [illegible] scientific terminology.

structure).
- Cataphoric Reference (Forward-Looking Specification):
- The referent's identity is established by a modifier that immediately follows the noun phrase (e.g., prepositional phrases, relative clauses):
- Text: "The concentration of the solute in the test tube was monitored." (Identified by 'of the solute').
- Text: "The vector that was transformed into E. coli contained kanamycin resistance." (Identified by the relative clause).
- Text: "The hypothesis that retroviruses cause oncogenesis gained widespread clinical backing."
- Exophoric Reference (Situational / Shared World Knowledge):
- Refers to entities universally known to exist uniquely, or entities immediately obvious in the physical room/environment:
- Text: "The Sun emits ionizing radiation", "The atmosphere", "The laboratory technician calibrated the hood".
- Shared Laboratory Context: "Please autoclave the waste canisters" (Both speaker and listener know which canisters are in the cleanroom).

>> Solution to Question 3:

[Problem Statement]: Detail the rules governing Zero Article (omission of articles) with uncountable nouns, plural nouns, and academic disciplines.

[illegible]

- Example B: "We observed a few aberrations" (Cautious finding indicating minor chromosomal instability).

[illegible]

- (Note: Individual lakes take zero article: Lake Geneva, Lake Superior).
- Political States Incorporating Common Nouns (republic, kingdom, union, states):
- The United States, The United Kingdom, The Czech Republic, The Netherlands (historical plural).
- Academic Institutions with Inverted 'Of'-Genitives:
- The University of Cambridge vs. Cambridge University (Zero article).
- The Massachusetts Institute of Technology vs. Caltech (Zero article).

[illegible]

>> Solution to Question [illegible] scientific registers.

- 'A lot of', 'Lots of', 'Plenty of', 'Heaps of':
- Strictly informal; unsuitable for peer-reviewed journal articles.
- Informal: "A lot of biomass was lost."
- Formal: "A significant quantity of biomass was lost" or "Substantial biomass was lost."
- Informal: "There were plenty of antibodies left."
- Formal: "An abundant supply of monoclonal antibodies remained."

[illegible] in biotechnology.

[Detailed Step-by-Step Answer]:
1. Syntactic Reduction in Idiomatic Coordinate Pairs:
In fixed prepositional pairs and balanced coordinate binominals, articles are conventionally dropped to enhance rhythmic velocity:
- "from top to bottom" (Not *from the top to the bottom).
- "hand in hand".
- "day by day".
- "in vitro and in vivo".
- "step by step".

2. Telegraphic Ellipsis in Technical Tables and Flowcharts:
In laboratory equipment flowcharts, table column headers, and benchside checklist protocols, articles are deliberately omitted for informational compression:
- "Mix reagents. Spin tube. Discard supernatant. Dry pellet."
- While appropriate for quick-reference SOP cards, full academic prose in research manuscripts requires restoring articles: "Mix the reagents. Spin the tube. Discard the supernatant, and dry the pellet."

[illegible]

- Neither (Negative disjunctive - takes singular noun and singular verb):
- Determiner: "Neither kidney displayed fibrosis."
- Pronoun: "Neither of the two samples was degraded." (Formal singular verb).

[illegible]

Uncombined chemical elements, gases, and simple compounds function as mass nouns and take ZERO article in general chemical discourse:
- "Oxygen and hydrogen combine to form water."
- "Carbon is the backbone of organic chemistry."
- "Ethanol was used to precipitate nucleic acids."

2. Specific Chemical Ions and Functional Groups:
When referring to discrete structural chemical entities or ions, the definite article is required:
- "The hydroxyl group (-OH) participates in hydrogen bonding."
- "The phosphate backbone confers a net negative charge to DNA."
- "The sodium ion (Na+) is pumped against its concentration gradient."

3. Bounded Reagents:
- "Add 10 mL of ethanol." (Zero article).
- "The ethanol used in this step was chilled to -20 deg C." (Definite article).

>>Solution to Question 20: [illegible] orders of magnitude.

referenced. The verb is plural ('were').
- Editorial Error: Writing "The inventor and founder were awarded..." creates an ungrammatical clash between a singular nominal structure and a plural verb.

[illegible]

- "Some cells lysed, while others remained intact."
- "Two cultures grew well, but the others were contaminated."

[illegible]

- "a single-cell RNA sequencing protocol" (Determiner 'a' governs the head noun 'protocol').
- "the enzyme-linked immunosorbent assay" (Determiner 'the' governs 'assay').

- Error 2: Article Over-Use (Inserting 'the' before generic mass nouns):
- Flawed: "*The biotechnology is transforming the medicine."
- Correction: "Biotechnology is transforming medicine."
- Rule: Broad generic fields, natural processes, and mass concepts take zero article.
